CSI 5340 Assignment4

Name: Lingjia Liu

Student Number: [redacted]

This report covers some implementations about the architecture of VAE, GAN, WGAN on MNIST and CIFAR10, some comparisons and analysis of model behaviors while modifying two key hyper-parameters (model complexity and latent dimension). In addition, exhibiting a group of generated images for each model and each dataset. This experiment is influenced by the virtual GPU, time, and units' restriction in COLAB.

1. VAE

1.1 MNIST

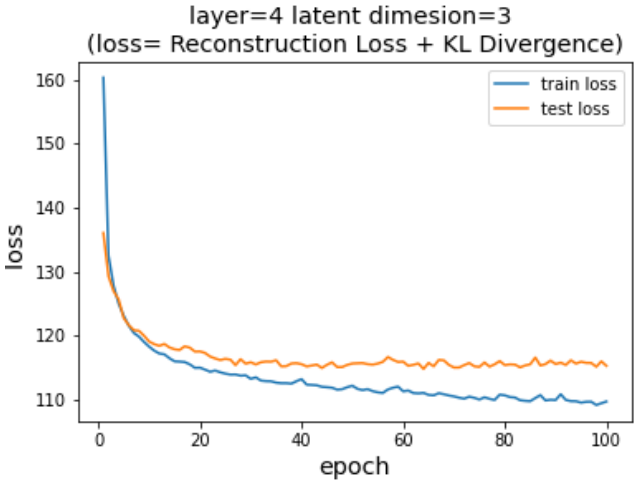
Training model with 4 linear encoding and decoding layers and using different latent dimensions (3 and 64) to compare, then changing the numbers of linear layers to 2 making another comparison.

- The relevant parameter setting:

	Model1	Model2	Model3
Encoding layers	4 linear	4 linear	2 linear
Decoding layers	4 linear	4 linear	2 linear
Latent dimensions	3	64	64
Optimizer	Adam		
Reconstruction loss	Binary cross-entropy		
Training loss	Reconstruction loss + Kl loss		

- Generated images and Loss Variation during training and testing.

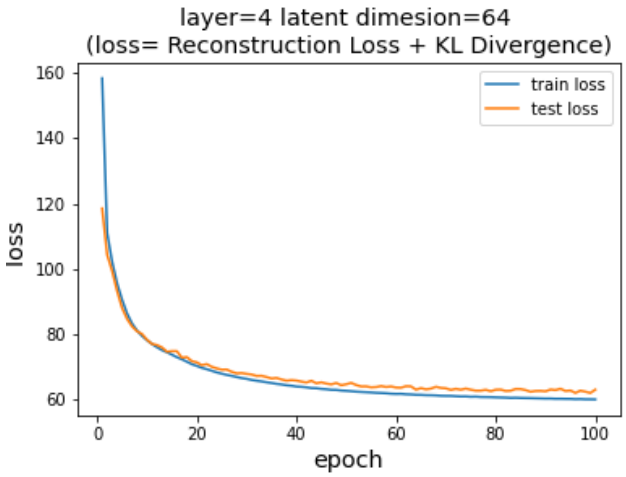




(a)generated image after 100 epochs *(b)loss variation*

Figure1.1.1: 4Layer 3dimesion

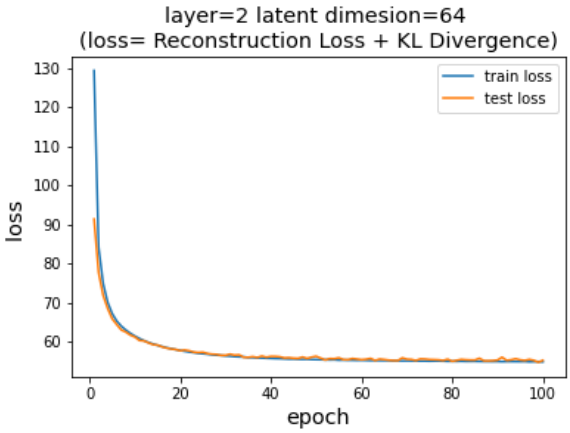




(a)generated image after 100 epochs *(b)loss variation*

Figure1.1.2: 4layer 64dimension





(a)generated image after 100 epochs *(b)loss variation*

Figure1.1.3: 2layer 64dimension

Figure 1.1.1 and 1.1.2 (ly4_dim64 VS ly4_dim3) shows that the training process of high dimension model is smoother than that of low dimension model, and generated images in the later model is clearer.

It is surprised that generated images in Figure1.1.2 and 1.1.3 are clearly, but it does not the original image or the code matters because of the increase in sharp degree with training epochs by observation. I also generate a random z and directly put into decode function to generate image(shown below).



ly4_dim3 ly4_dim64 ly2_dim64

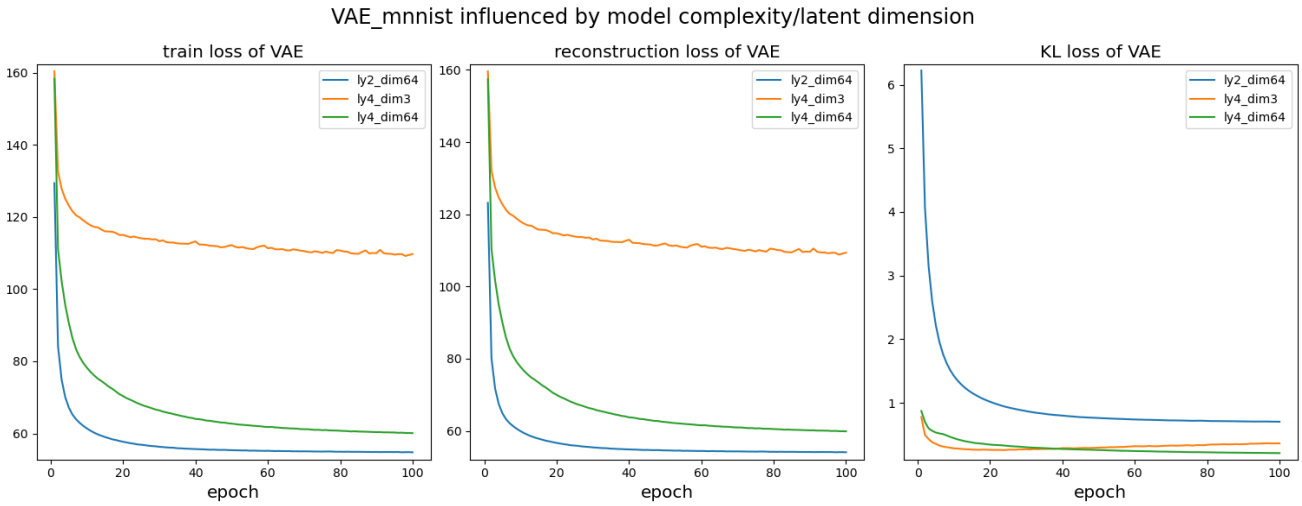


Figure 1.1.4: comparison

The comparisons in Figure 1.1.4 illustrates that, whether at start or in the end, training loss is high in the model with small latent dimension (ly4_dim64 VS ly4_dim3). Although this value of model with simple complexity is lower than that with complex one (ly4_dim64 VS ly2_dim64), the difference is far less significant. This is true for the reconstruction loss, though presents a totally opposite picture for KL loss.

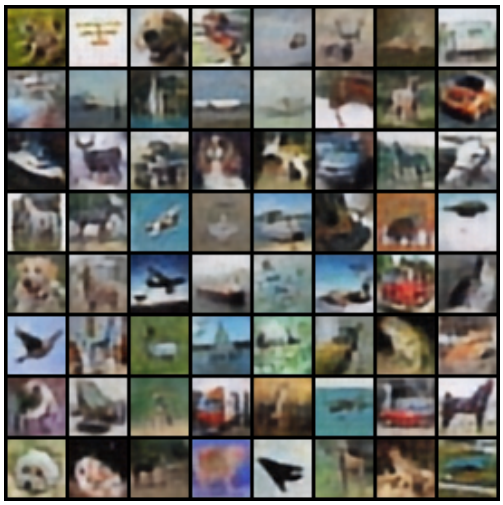
1.2 CIFAR

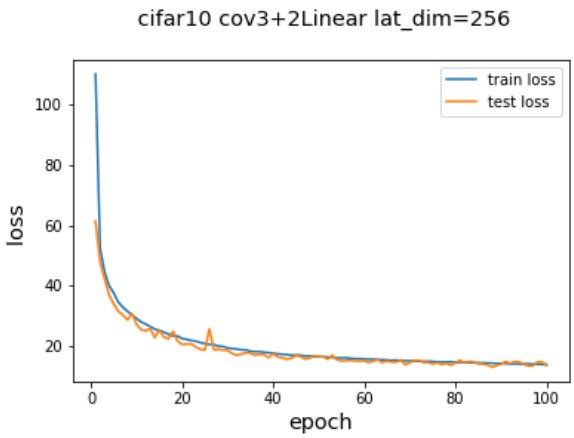
Training model with 3 convolutional layers 2 linear layers of encoding and decoding layers and using different latent dimensions (256 and 512) to compare, then changing the numbers of linear layers to 3 making another comparison.

- The relevant parameter setting:

	Model1	**Model2**	**Model3**
Encoding layers	3conv+2linear	3conv+2linear	3conv+3linear
Decoding layers	3conv+2linear	3conv+2linear	3conv+3linear
Latent dimensions	256	512	512
Optimizer	Adam		
Reconstruction loss	Mean Squared Error		
Training loss	Reconstruction loss + Kl loss		

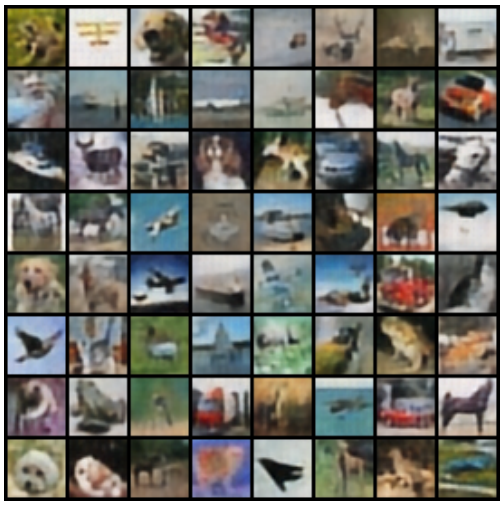
- Generated images and Loss Variation during training and testing.

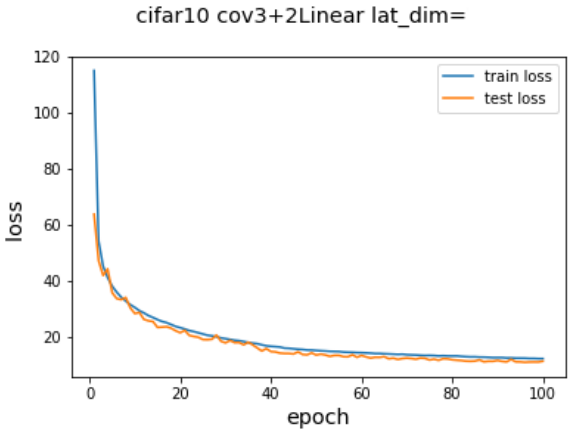




(a)generated image after 100 epochs *(b)loss variation*

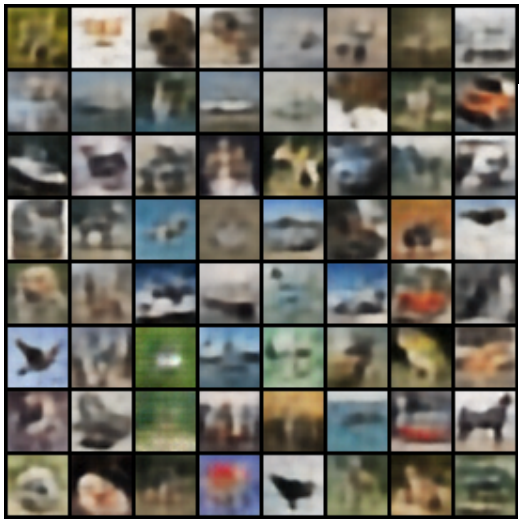
Figure 1.2.1: 3conv+2linear 256 dimension

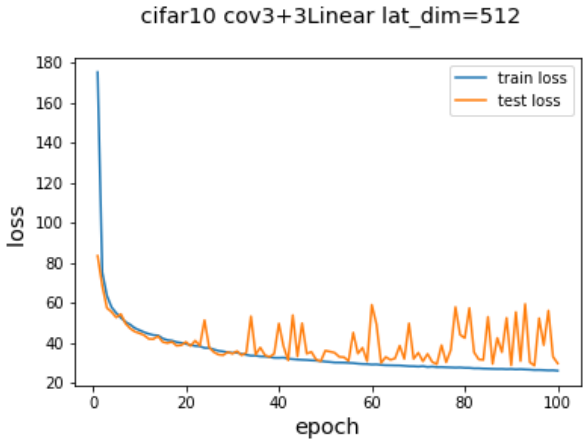




(a)generated image after 100 epochs *(b)loss variation*

Figure 1.2.2: 3conv+2linear 512 dimension





(a)generated image after 100 epochs *(b)loss variation*

Figure 1.2.3: 3conv+3linear 512dimension

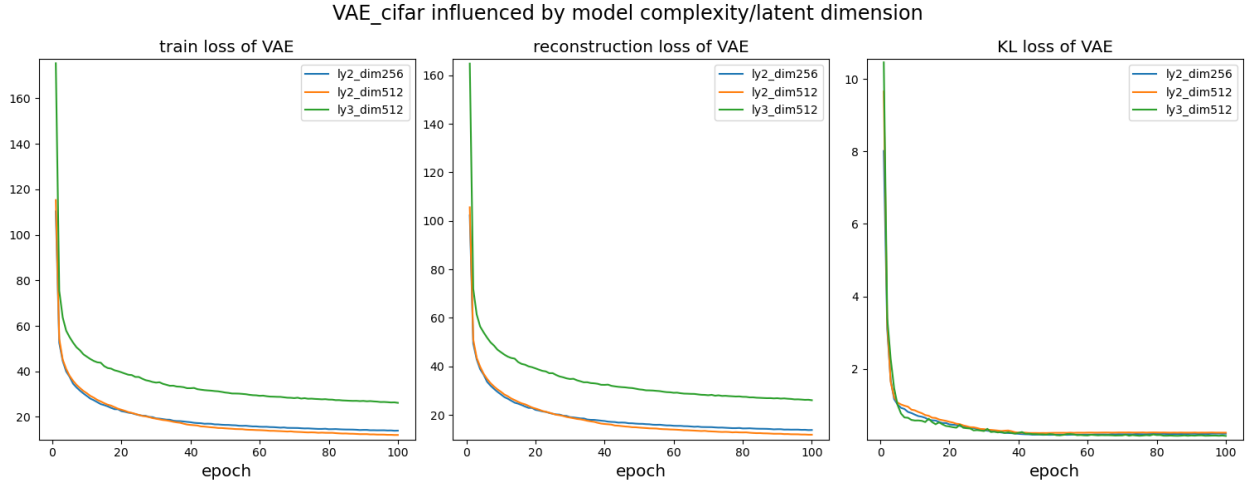


Figure 1.2.4: comparison

According to figure1.2.4, the change in latent dimension (ly2_dim256 VS ly2_dim512) exerts an unobvious variation of train loss, whereas the rise in network layer (ly2_dim512 VS ly3_dim512) enables training loss to increase not only at beginning but also during the whole process, which is the same in reconstruction loss. Furthermore, KL loss in all of them are large at start and then dramatically decrease to nearly 0.\

2. GAN

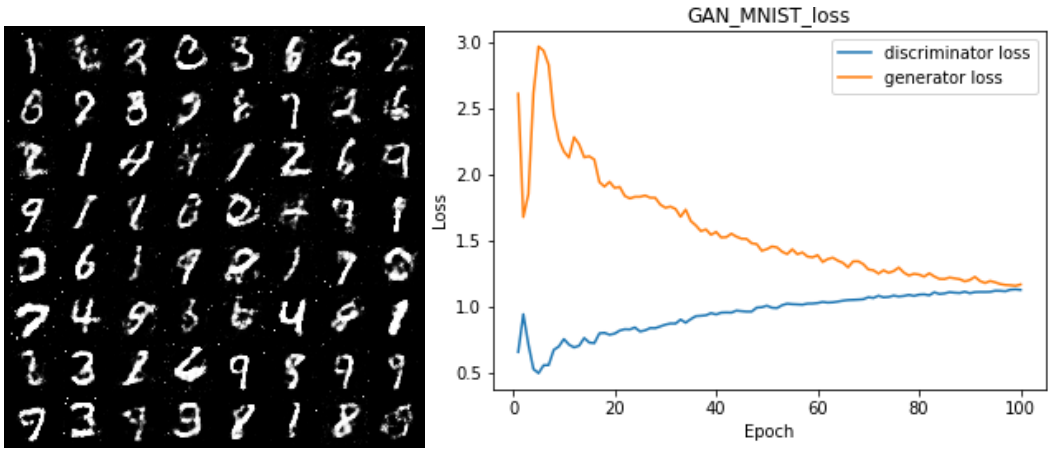
2.1 MNIST

Training model with 3 linear of generator and discriminator layers and using different latent dimensions (64 and 128) to compare, then changing the numbers of linear layers to 4 making another comparison.

- The relevant parameter setting:

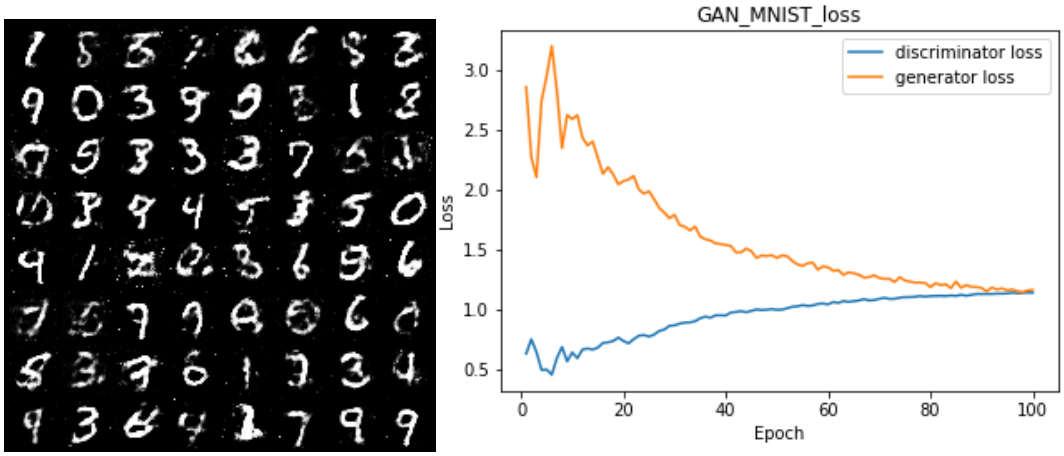
	Model1	Model2	Model3
Generator layers	3linear	3linear	4linear
Discriminator layers	3linear	3linear	4linear
Latent dimensions	128	64	64
Activation	LeakyRelu0.2		
Dropout(discriminator)	0.3		
Loss criterion	Binary cross-entropy		
Optimizer	Adam		

- Generated images and Loss Variation during training and testing.



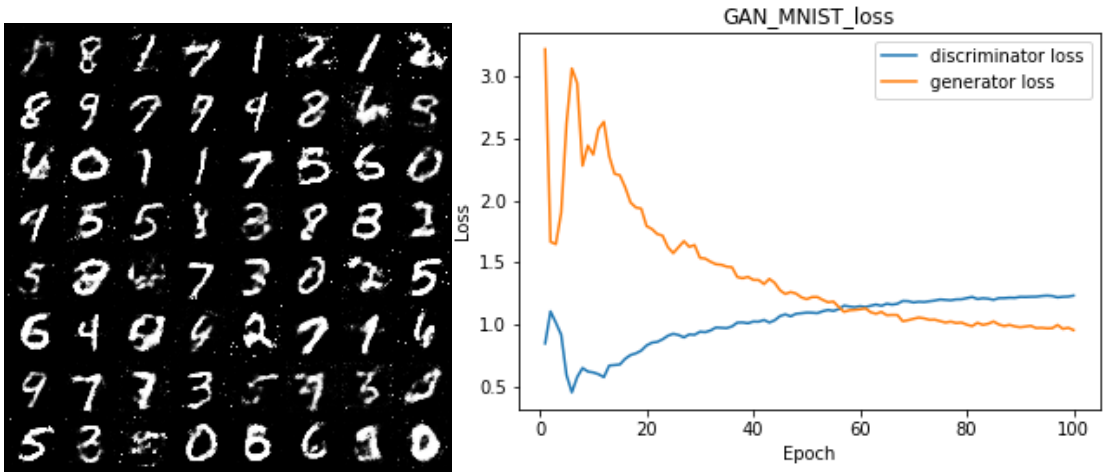
(a)generated image after 100 epochs *(b)loss variation*

Figure 2.1.1: 3 linear 128dimension



(a)generated image after 100 epochs *(b)loss variation*

Figure 2.1.2: 3 linear 64dimension



(a)generated image after 100 epochs *(b)loss variation*

Figure 2.1.3: 4 linear 64dimension

There is no obvious difference in the discriminator loss and generator loss between ly3_dim128 and ly3_dim64 (Figure 2.1.1 and 2.1.2) except the former seems to need more iterations to converge. Moreover, Figure 2.1.2 and 2.1.3 illustrates that the discriminator loss and generator loss in layer4_ dim64 fluctuate more dramatically at the beginning and converge earlier than that in layer3_dim64, but the generated images are sharper.

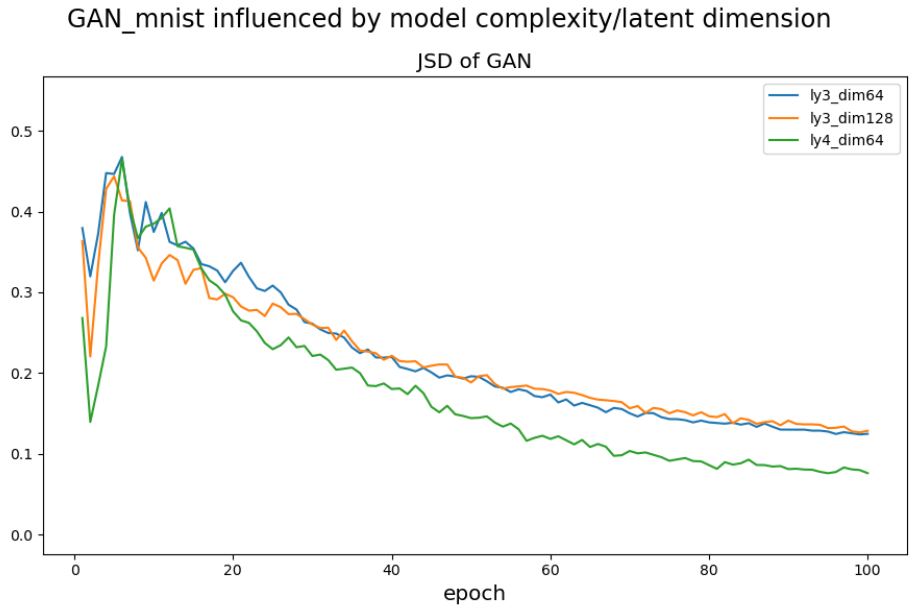


Figure 2.1.4: comparison

According to Figure2.1.4, the JSD of the three models experiences a fluctuated decline before increasing obviously. To be noticed, although JSD of Ly3_dim64 is higher than that of ly3_dim128 at an early stage, after which the difference of decline trend is far less significant (to almost 0.15) and the decline trend of ly4_dim64 is more obvious than ly3_dim64, with the lowest JSD to less than 0.1.

2.2 CIFAR

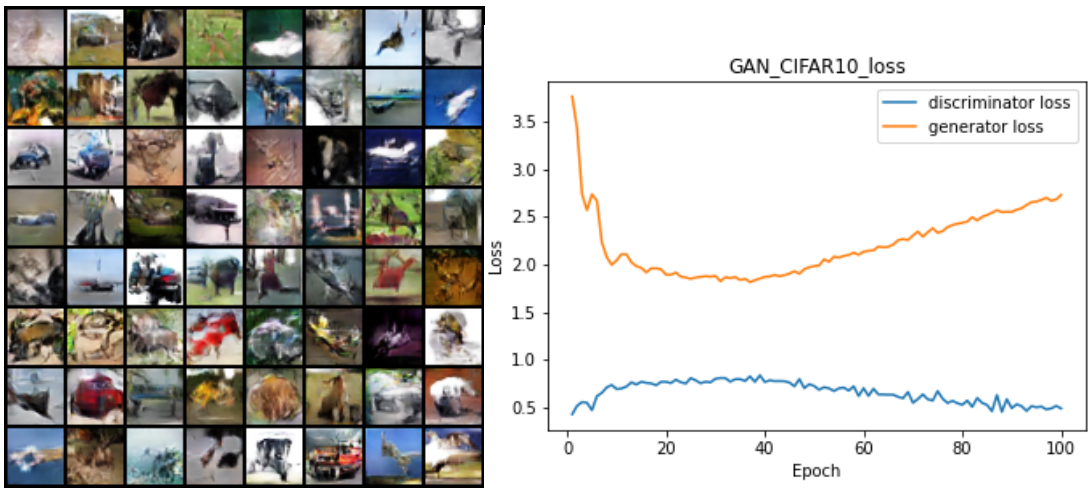
Training model with 4 convolutional layers of generator and discriminator layers and using different latent dimensions (64 and 128) to compare, then changing the numbers of layers to 5 making another comparison.

- The relevant parameter setting:

	Model1	Model2	Model3
Generator layers	4conv	4conv	5conv

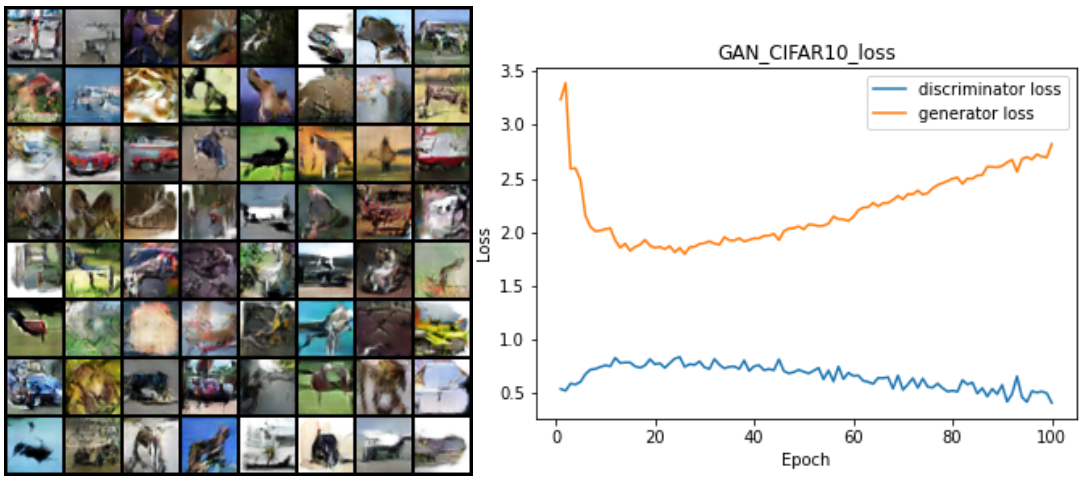
Discriminator layers	4conv	4conv	5conv
Latent dimensions	128	64	64
Generator Activation	Relu		
Discriminator Activation	LeakyRelu0.2		
Loss criterion	Binary cross-entropy		
Optimizer	Adam		

- Generated images and Loss Variation during training and testing.



(a)generated image after 100 epochs *(b)loss variation*

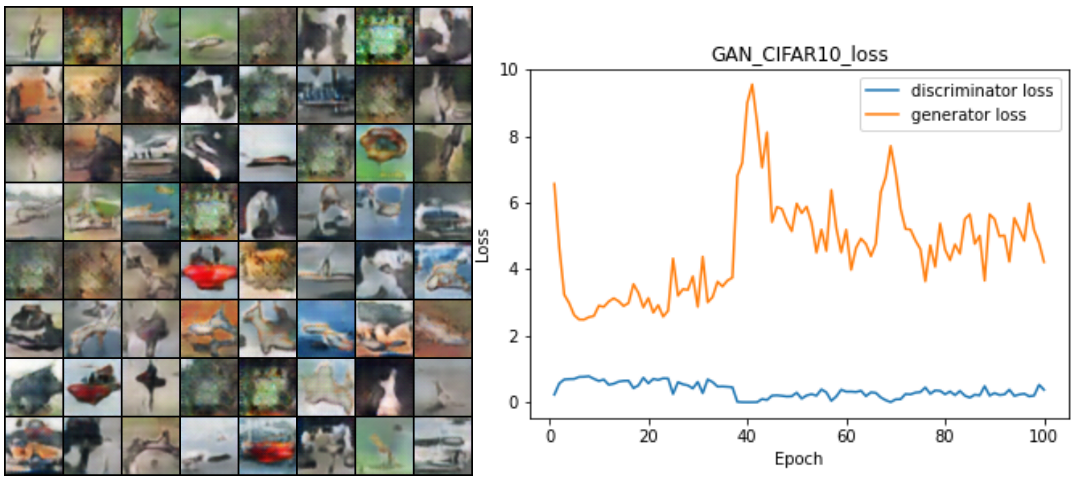
Figure 2.2.1: 4conv 128dimension



(a)generated image after 100 epochs *(b)loss variation*

Figure 2.2.2: 4conv 64dimension

Comparing figure 2.2.1(b) and 2.2.2(b), figure 2.2.2(b) and 2.2.3(b), both absolute value of generator loss and critic loss in ly4_dim64, are lower than that in ly4_dim128 and ly5_dim64. Noticeably, differences between ly4_dim64 and ly4_dim128 are far less significant. Instead, the generator loss in ly4_dim64 is obvious lower than that in ly5_dim64.



(a)generated image after 100 epochs *(b)loss variation*

Figure 2.2.3: 5conv 64dimension

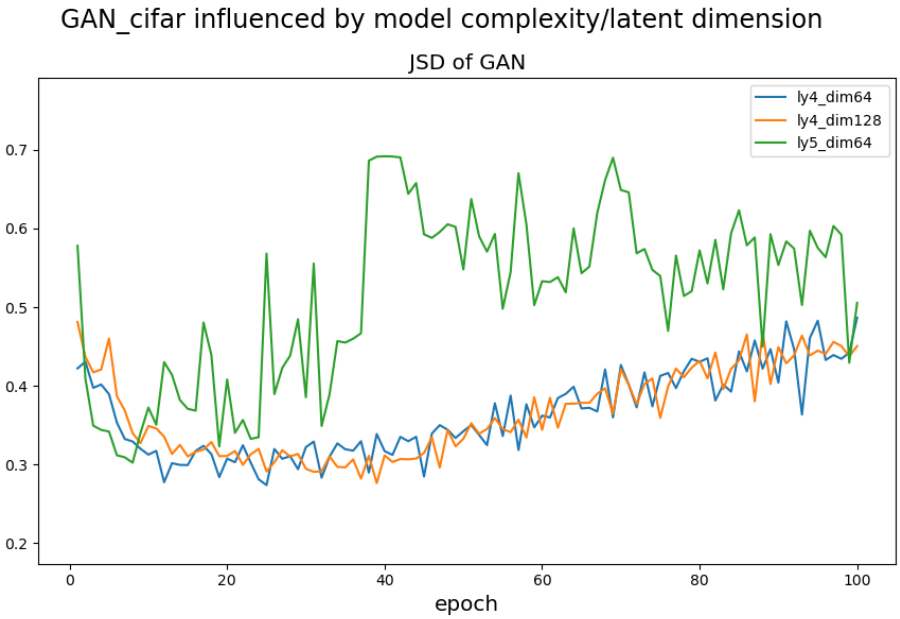
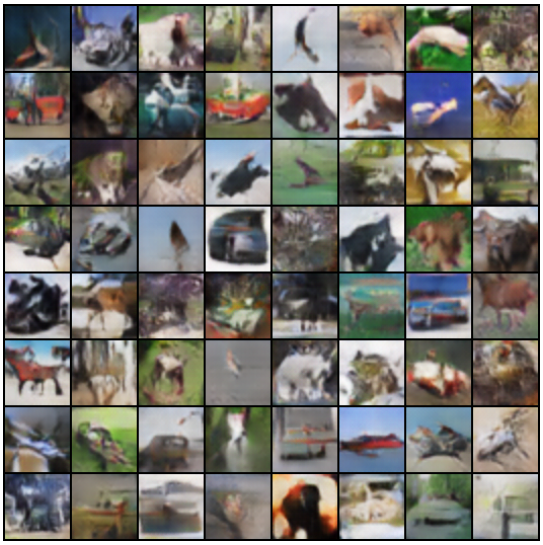


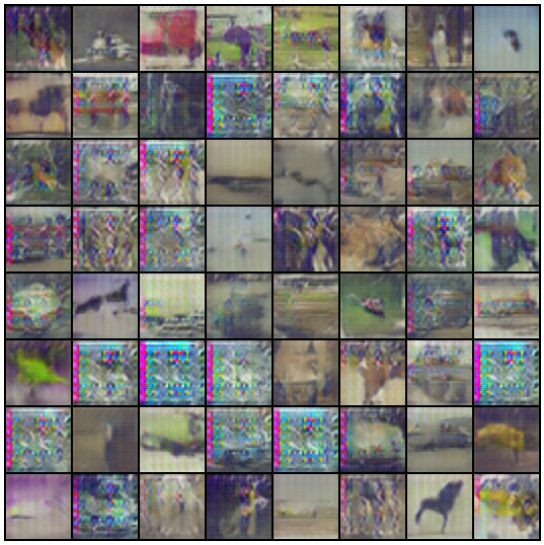
Figure 2.2.4: comparison

JSD for GAN on cifar10 decline at first and then present an overall increased trend with fluctuation, in Figure2.2.4, to be specific, the trends for ly4_dim128 and ly4_dim64 show a similar picture expect the former is a little higher than the latter at start. Surprisingly, the JSD variation of ly5_dim64 is dramatic between 0.3 and 0.7 which is higher than ly4_dim64 during the whole iterations.

As for figure 2.2.3, generator loss is 4 times higher than discriminator loss or even more than that, which makes me to train this model again and then it appears model collapse, shown in figure 2.2.5.



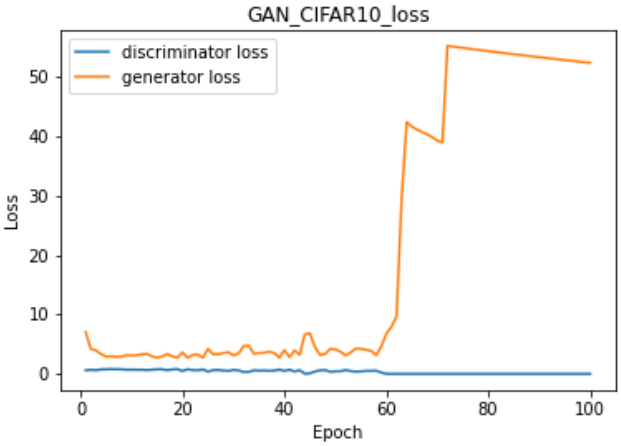
(a)generated image 58 epoch



(b) generated image 59 epoch



(c) generated image 100 epoch



(d)loss variation

Figure 2.2.5: model collapse after 59 epochs

3. WGAN

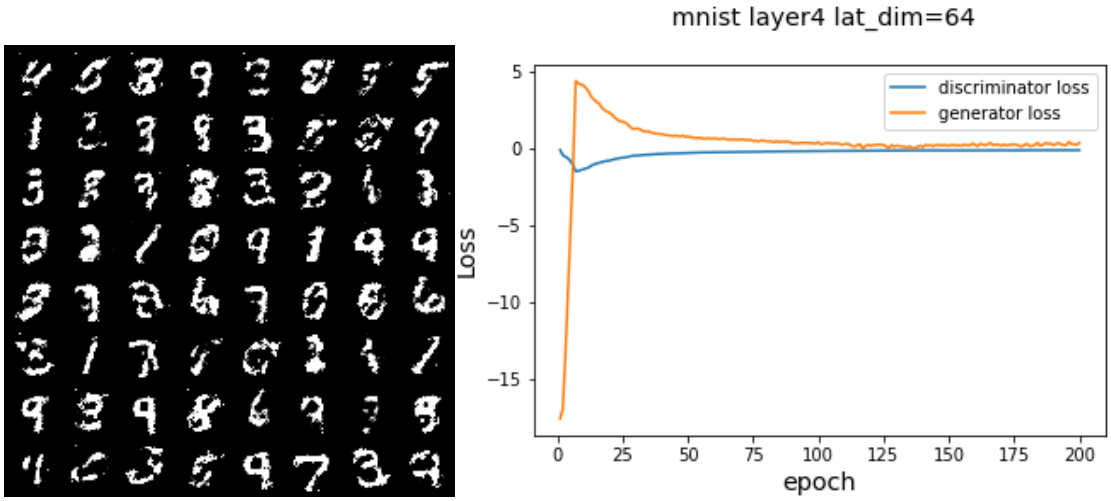
3.1 MNIST

Training model with 4 linear of generator and discriminator layers and using different latent dimensions (64 and 128) to compare, then changing the linear layers to convolutional layers to make another comparison.

- The relevant parameter setting:

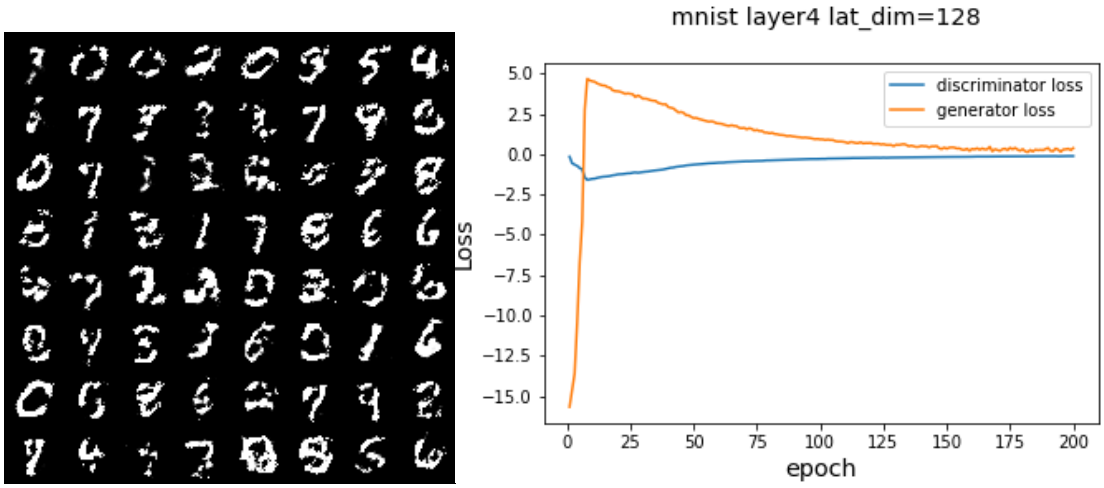
	Model1	Model2	Model3
Generator layers	4linear	4linear	4conv
Critic layers	4linear	4linear	4conv
Latent dimensions	64	128	128
Optimizer		RMSprop	
Weight clipping		0.01	

- Generated images and Loss Variation during training and testing.



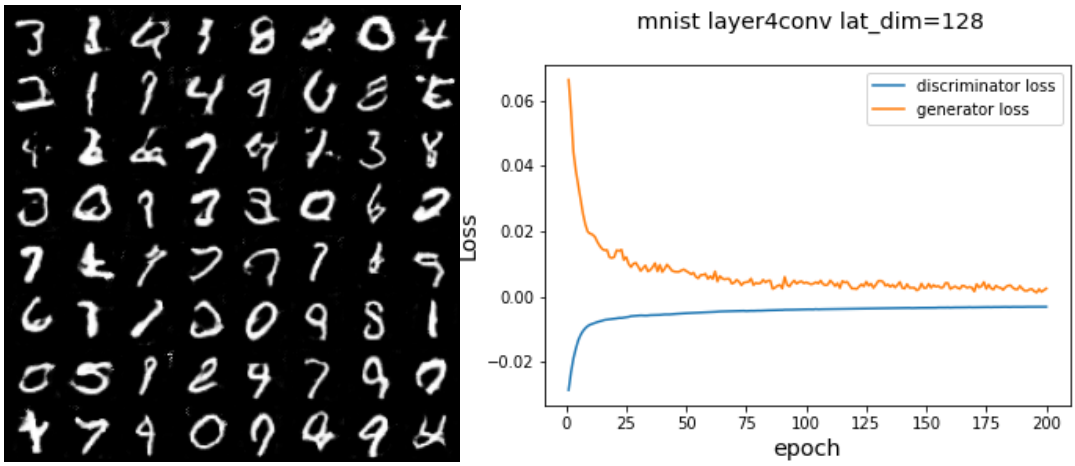
(a)generated image after 200 epochs *(b)loss variation*

Figure 3.1.1: 4linear 64dimension



(a)generated image after 200 epochs *(b)loss variation*

Figure 3.1.2: 4linear 128dimension



(a)generated image after 200 epochs *(b)loss variation*

Figure 3.1.3: 4conv 128dimension

Figure 3.1.1 and 3.1.2 shows the difference between ly4_dim64 and ly4_dim128, the generator and critic loss for the former converge faster than the later one.

In addition, it is apparent that the generated images of model using 4 convolutional layers(ly4conv_dim128) performs higher quality than model using 4 linear layers, which means the performance of ly4conv_dim128 may be enhanced more with more epoch.

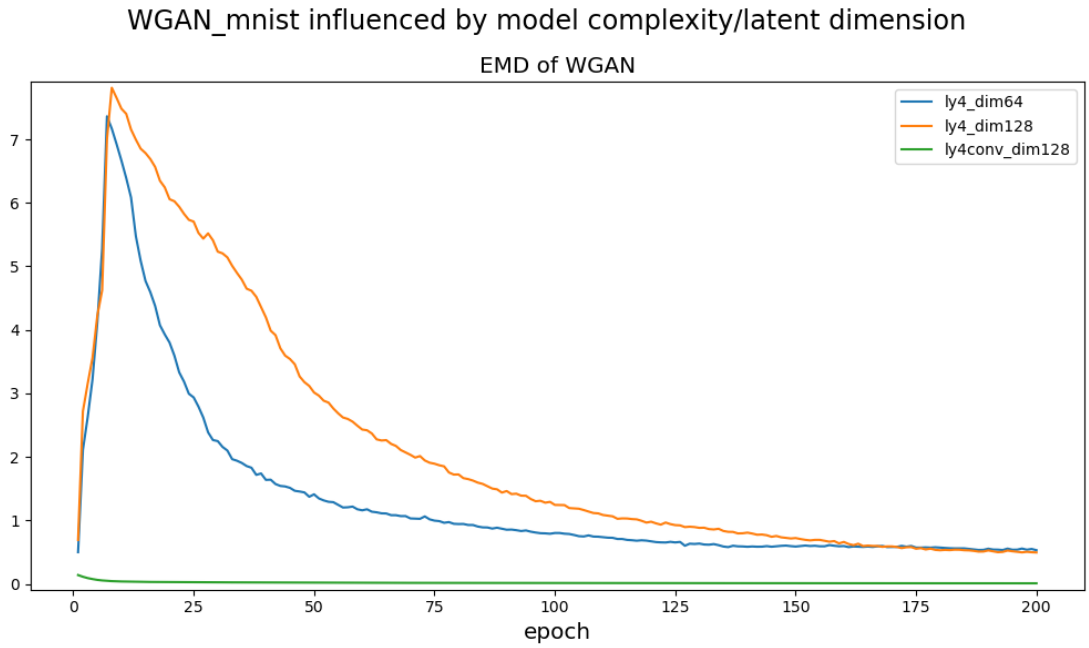


Figure 3.1.4: comparison

In figure3.1.4, although the EMD for the model with 4 linear layers (ly4_dim64, ly4_dim128) gradually decline to 0.5 after peaking at more than 7, the value is still dramatic larger than the model with 4 convolutional layers(ly4conv_dim128), whose EMD is nearly 0.1 with each epoch. In addition, EMD decrease earlier to low level (EMD=1) in ly4_dim64 than that in ly4_dim128.

3.2 CIFAR

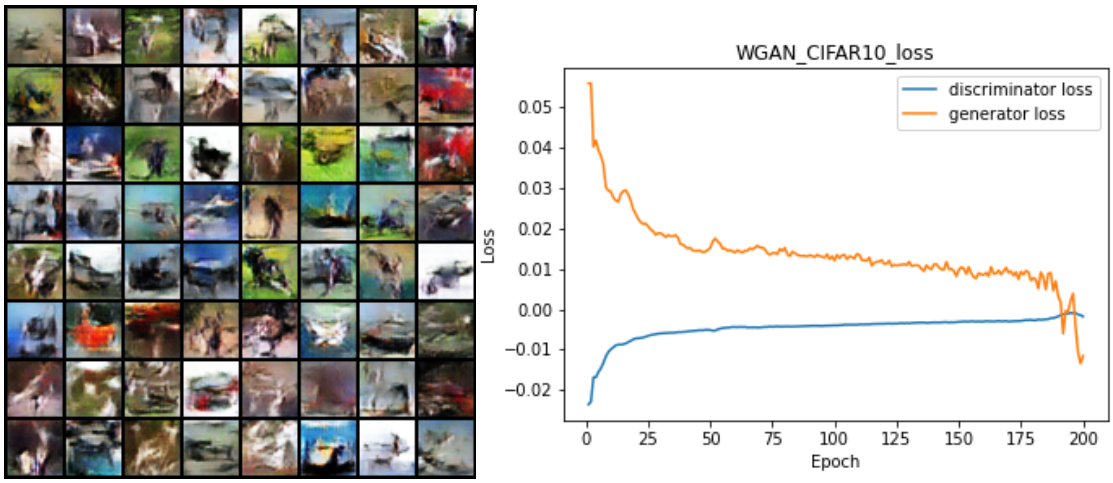
Training model with 4 convolutional layers of generator and critic layers and using different latent dimensions (64 and 128) to compare, then changing the numbers of layers to 5 making another comparison.

- The relevant parameter setting:

	Model1	Model2	Model3
Generator layers	4conv	4conv	5conv
Critic layers	4conv	4conv	5conv
Latent dimensions	128	64	64
Generator Activation	Relu		
Discriminator Activation	LeakyRelu0.2		

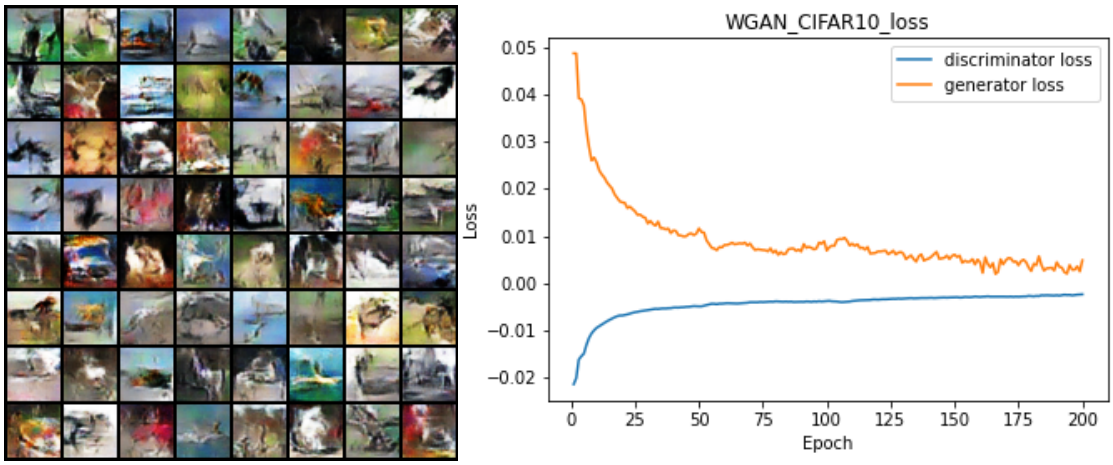
Optimizer	RMSprop
Weight clipping	0.01

- Generated images and Loss Variation during training and testing.



(a)generated image after 200 epochs *(b)loss variation*

Figure3.2.1: 4conv 128dimension

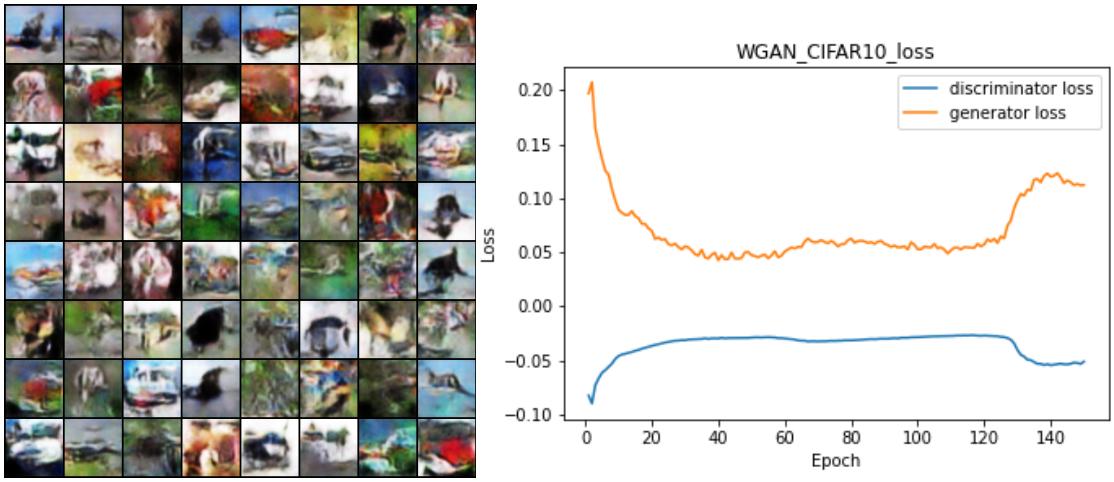


(a)generated image after 200 epochs *(b)loss variation*

Figure 3.2.2: 4conv 64dimension

Comparing figure 3.2.1(b) and 3.2.2(b), figure 3.2.2(b) and 3.2.3(b), both absolute value of generator loss and critic loss in ly4_dim64, are lower than that in ly4_dim128 and ly5_dim64. To be noticed, differences between ly4_dim64 and ly4_dim128 are far

less significant. Instead, the generator loss in ly4_dim64 is obvious lower than that in ly5_dim64.



(a)generated image after 200 epochs *(b)loss variation*

Figure 3.2.3: 5conv 64dimension

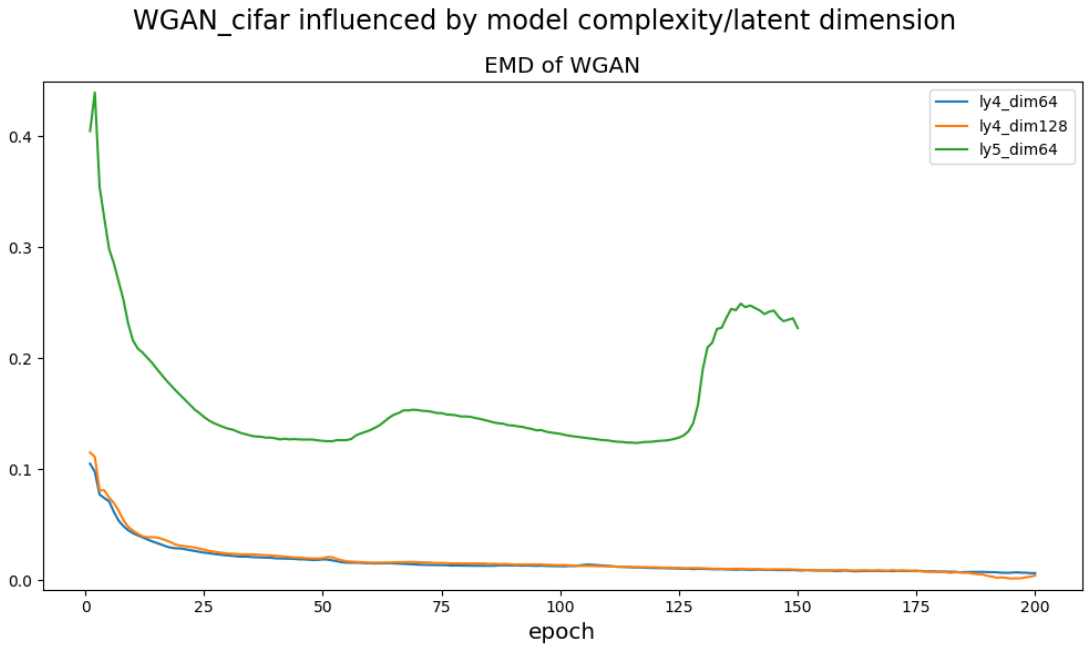


Figure 3.2.4: comparison

It can be seen that, in the figure 3.2.4, EMD of model with the ly4_dim64 and ly4_dim128 experience highly similar trend declining from 0.1 to almost 0. Surprisingly, the EMD for ly5_dim64, which keeps the largest number within 150 epochs, presents a slight variation after declining from 0.43 to 0.15, and then dramatically climbs by 0.1 to 0.25. (Because of the unit's limitation of COLAB, the model3 only trained with 150 epochs.)